SHAP-σ: A Label-Free Rashomon Pipeline for Multivariate Time-Series Anomaly Detection

Ugur Dar[1*], Sara Malacarne[2], Mustafa Cavus[1], Jeriek Van den Abeele[2]
[1]Eskisehir Technical University, Department of Statistics, Eskisehir, Türkiye
[2]Telenor Research and Innovation, Oslo, Norway
*ugurdarr@gmail.com

***Abstract*—We address unsupervised anomaly detection on multivariate cellular network KPI streams, where labels are unavailable for both training and model selection. We propose SHAP-σ, a label-free pipeline that self-tunes a pool of $K{=}10$ unsupervised detectors, selects a small *Rashomon set* by a label-free joint quality $\times$ diversity score (Mass–Volume quality and score-decorrelation; closed-form at the pair level), and gates each anchor flag by the cross-detector standard deviation of TreeSHAP attributions – promoting attribution disagreement from a post-hoc explanation into a per-KPI evidence signal. On TELCO (12 per-KPI label streams), a label-free variance-stability rule selects a five-detector Rashomon set reaching per-KPI macro $F_1{=}0.460$, above the best self-tuned singleton (0.419) and well above the deep baselines DC-VAE (0.134 [1]) and C-MTAD-GAT (0.160 [2]) reported under the same per-KPI aggregation, and stays in the 0.45–0.46 band for every set size $K_R \geq 3$ (no label-based tuning). The gain over singletons stems chiefly from the label-free self-tuning and subset selection; the σ gate matches a naive ensemble vote on F_1 but is more stable across seeds and, unlike a black-box vote, yields explicit per-KPI attribution evidence – a transparent, reproducible pipeline throughout.**

***Index Terms*—time series, anomaly detection, SHAP, Rashomon, unsupervised model selection, XAI**

I. Introduction

Unsupervised anomaly detection in multivariate time series underpins fault diagnosis in domains as diverse as cellular networks, server fleets, and industrial telemetry. Yet a persistent obstacle hinders deployment: without ground-truth labels, the practitioner cannot reliably decide which detector to trust. Internal validation metrics that work well for clustering rarely transfer to anomaly detection, and recent large-scale empirical work shows that most label-free criteria perform near random when ranking modern detectors [3].

This selection problem is sharpened by the *Rashomon effect* [4]: for any non-trivial task, a multiplicity of near-optimal models exists, each making different errors [5], [6]. We argue that this attribution disagreement is not noise to be averaged away but a signal to be exploited: at a flagged point, the spread of attributions across the Rashomon set indicates that at least one detector commits to a feature-level explanation, providing second-source evidence beyond the anchor's score.

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Code and data for full reproducibility: `https://github.com/ugurdar/shap_sigma_paper`.

Prior label-free model selection has explored three avenues. Excess-Mass and Mass-Volume curves [7] give geometry-based scores but assume well-behaved density estimates. Meta-learning approaches such as MetaOD [8] and ELECT [9] transfer historical-performance signals across datasets, but require an offline labelled meta-corpus. Most recently, robust Borda rank aggregation across multiple imperfect internal metrics has been proposed for time-series AD [10]; the large-scale study of [3] shows that no internal metric universally dominates. PyOD [11] has standardized detector implementations. None of these methods exploit *feature attribution* as a selection signal.

Independently, SHAP [12], [13] has become the de facto post-hoc explainer for anomaly detection, mostly to explain a single chosen model; the closest precedent is Levy et al. [14], who use SHAP attributions for ensemble member *selection* in supervised tabular AD, whereas we use cross-detector SHAP standard deviation as an attribution-evidence *gate* in the unsupervised TS regime. Modern deep TS-AD methods [15]–[18] likewise commit to a single architecture. Outlier ensemble theory [19], [20] and the classical diversity literature [21] establish that complementary errors improve ensemble accuracy, yet the standard diversity measures (Q-statistic, disagreement) require labels. Following [20], we replace label-dependent diversity with the Spearman rank correlation of training scores, which [20] identifies as the practical label-free proxy in OD ensembles.

a) Contributions.: (1) Label-free self-tuning of the full $K{=}10$ detector pool (PCA via 95% explained variance; k-NN/LOF via k-distance elbow; GMM via BIC; HBOS via Sturges' rule; an AutoEncoder and OCSVM alongside the eight PyOD detectors), which raises detector-level F_1 without using any anomaly labels. (2) A closed-form Rashomon pair score $s_{ij} = \frac{1}{2}(q_i + q_j)(1 - \rho_{ij})$, where q_d is a label-free Mass–Volume quality proxy, jointly rewarding label-free quality and inter-detector score decorrelation. (3) A σ-SHAP attribution-evidence filter using cross-detector SHAP standard deviation as a second-source signal that corroborates anchor flags on per-KPI labels. (4) Empirical demonstration on the TELCO cellular benchmark that the label-free pipeline exceeds the best self-tuned singleton and the reported deep baselines DC-VAE and C-MTAD-GAT [1], [2].

b) Literature gap.: To our knowledge, no published method combines (a) label-free detector selection with (b)

SHAP attributions as a *filter* signal rather than a post-hoc explanation. Rashomon-set analysis has so far been developed in supervised regimes [5], [6]; its extension to unsupervised anomaly detection is open. Robust rank aggregation [10] pools imperfect label-free metrics but never inspects attributions, while [3] diagnoses their weakness without proposing a disagreement-based pipeline. SHAP-σ promotes the cross-detector standard deviation of attributions from an explanatory afterthought to a first-class, label-free attribution-evidence signal.

The remainder of the paper is organized as follows. Section II formalizes joint pair scoring and the σ-SHAP filter. Section III describes datasets and protocol. Section IV reports empirical results. Section V concludes.

II. Methodology

We propose **SHAP-σ**, a label-free pipeline that selects a small subset $\mathcal{R} \subseteq \mathcal{D}$ of detectors from a pool $\mathcal{D}$ by jointly optimising label-free quality and pairwise diversity, explains every member of $\mathcal{R}$ with a tree surrogate, takes TreeSHAP attributions [12], [13], and converts the cross-detector spread of those attributions into a per-point, per-feature attribution-evidence gate. Labels are used nowhere in training, ranking, or set selection.

A. Notation

Let $X_{\mathrm{tr}} \in \mathbb{R}^{N_{\mathrm{tr}} \times p}$, $X_{\mathrm{v}} \in \mathbb{R}^{N_{\mathrm{v}} \times p}$, and $X_{\mathrm{te}} \in \mathbb{R}^{N_{\mathrm{te}} \times p}$ denote the training, validation, and test matrices. The detector pool is $\mathcal{D} = \{d_1, \ldots, d_K\}$ with $K{=}10$. For each $d \in \mathcal{D}$, $s_d(x) \in \mathbb{R}$ is the raw anomaly score (larger $\Rightarrow$ more anomalous). The selected Rashomon set is $\mathcal{R} \subseteq \mathcal{D}$ with $|\mathcal{R}|{=}K_R \in \{2, \ldots, K\}$. We write $a \in \mathcal{R}$ for the anchor.

B. Detector Pool and Label-Free Self-Tuning

$\mathcal{D}$ consists of $K{=}10$ heterogeneous unsupervised detectors: the 8 PyOD [11] detectors used in prior work – ECOD [22], COPOD [23], HBOS [24], k-NN [25], LOF [26], PCA [27], GMM density score [19], IF [28] – together with OCSVM [29] and a PyTorch AutoEncoder. The pool deliberately mixes tree-, density-, distance-, projection-, kernel-, and neural inductive biases, which is what makes the cross-detector σ in Stage 4 informative.

a) Label-free self-tuning.: PyOD default hyperparameters can fail silently – e.g. PCA with `n_components=None` collapses to constant-zero scores on some inputs. We therefore tune every parameterised detector *label-free* using only X_{tr}:

- **PCA**: `n_components` chosen so that cumulative explained variance is $\geq 95\%$ on X_{tr}.
- **k-NN and LOF**: `n_neighbors` from the k-distance elbow [30] – the rank that maximises the deviation of the mean k-th-nearest-neighbour distance from the diagonal of its normalised cumulative curve.
- **GMM**: `n_components` minimising the Bayesian Information Criterion (BIC) over $\{1, \ldots, 10\}$ components.
- **HBOS**: `n_bins` from Sturges' rule, $\lfloor \log_2 N_{\mathrm{tr}} \rfloor + 1$.
- **IF**: `n_estimators` from a path-length-std plateau (first grid point with $< 2\%$ relative change).
- **AutoEncoder**: fixed $[64, 32]$ encoder trained for 50 epochs with batch size 64.
- **OCSVM**: RBF kernel with $\nu{=}0.05$; **ECOD, COPOD**: parameter-free.

Each detector exposes a single hyperparameter we tune; the choices are deterministic functions of X_{tr} and therefore use no test label whatsoever. The effect on TELCO is documented in Sec. IV-A.

C. Stage 1 – Label-Free Quality via a Mass–Volume Proxy

We need a continuous label-free quality score $q_d \in [0, 1]$. Following the Mass–Volume principle of Goix [7] – a selective detector concentrates its high scores on training mass and spreads less of them over a uniform background – we use a fast, surrogate-based single-level MV proxy that preserves the ranking-relevant structure while avoiding full curve integration. For each d, we fit a LightGBM surrogate $\hat{s}_d$ on $(X_{\mathrm{tr}}, s_d(X_{\mathrm{tr}}))$ and define

$$A_d \;=\; \Pr_{X_{\mathrm{u}}}\big[\hat{s}_d(X_{\mathrm{u}}) \;>\; Q_{0.9}\big(s_d(X_{\mathrm{tr}})\big)\big], \tag{1}$$

the surrogate-predicted background tail mass above the empirical 0.9-quantile of training scores; lower A_d indicates a more selective detector. We then min–max invert,

$$q_d \;=\; \frac{\max_e A_e - A_d}{\max_e A_e - \min_e A_e} \;\in\; [0, 1], \tag{2}$$

estimating Eq. (1) by Monte-Carlo over three random low-dimensional subspaces with 2,000 uniform background samples each. We use a single-level MV proxy rather than the full EM/MV average [7]: [3] reports that combining imperfect internal metrics need not improve ranking, and on TELCO Eq. (1) ranks LOF – the best self-tuned singleton ($F_1{=}0.419$) – highest. A full Goix EM/MV-AUC ablation is deferred to a journal extension.

D. Stage 2 – Joint Pair Scoring and the Rashomon Set

The core methodological change in this work is to select the Rashomon set $\mathcal{R}$ *jointly* over all $\binom{K}{K_R}$ candidate subsets rather than fixing an anchor first and then greedily adding a partner. Joint selection makes the pipeline a Rashomon procedure in the sense of [5], [6]; the closest supervised precedent is the selective-ensemble construction of Black et al. [31], which we adapt to the unsupervised setting by replacing label-dependent agreement with Spearman score correlation.

a) Pairwise diversity.: Let $\rho_{ij} = \mathrm{Spearman}(s_i(X_{\mathrm{tr}}), s_j(X_{\mathrm{tr}}))$. Diversity is $1 - \rho_{ij} \in [0, 2]$; rank correlation is robust to the heterogeneous score scales of $\mathcal{D}$ [21].

b) Three label-free strategies.: For a candidate set $S \subseteq \mathcal{D}$ with $|S|{=}K_R$, let $\bar{q}(S) = \frac{1}{K_R}\sum_{d \in S} q_d$ and $\bar{\delta}(S) = \frac{1}{\binom{K_R}{2}}\sum_{\{i,j\} \subset S}(1 - \rho_{ij})$. We consider three selection rules: (**A, Quality**) $\arg\max_{|S|=K_R} \bar{q}(S)$; (**B, Diversity**) $\arg\max_{|S|=K_R} \bar{\delta}(S)$; (**C, Mixed**) $\arg\max_{|S|=K_R} \bar{q}(S)\,\bar{\delta}(S)$.

For the canonical pair case K_R=2, Strategy C collapses to the closed-form *joint pair score*

$$(i^*, j^*) \;=\; \arg\max_{1\le i<j\le K} \frac{q_i + q_j}{2}\big(1-\rho_{ij}\big), \tag{3}$$

which is the centrepiece of our method.

c) Anchor.: Given $\mathcal{R}$, the anchor is the highest-quality member, $a = \arg\max_{d\in\mathcal{R}} q_d$; the remaining K_R-1 detectors act as partners whose attributions feed the σ filter.

d) Label-free set size.: The closed-form pair (Eq. (3)) is the minimal K_R=2 case, but a cross-detector standard deviation estimated from only two members is statistically fragile (its value is fixed by a single difference). A reliable σ therefore *requires* a larger set, so we fix the Rashomon set label-free by a variance-stability rule: retain every detector whose quality exceeds the pool median, $\mathcal{R} = \{d : q_d > \mathrm{median}_e\, q_e\}$. This uses no labels and no K_R tuning; on TELCO it selects five detectors. Crucially the result is *insensitive* to this choice: the F_1 sweep over K_R (Sec. IV-B, Table II) plateaus at 0.45–0.46 for every $K_R \ge 3$, so no label-dependent size selection is needed. We expose $(K_R, \mathcal{S})$ as label-free hyperparameters following [3], [10], who argue that no internal criterion universally dominates.

E. Stage 3 – Surrogate SHAP Attribution

For each $d \in \mathcal{R}$ we fit a LightGBM regressor [32] g_d on $(X_{\mathrm{tr}}, s_d(X_{\mathrm{tr}}))$, compute exact TreeSHAP [12], [13] attributions $\Phi_d^{\mathrm{v}}, \Phi_d^{\mathrm{te}}$, and normalise each detector by its global validation standard deviation, $\tilde{\Phi}_d^{\bullet} = \Phi_d^{\bullet}/(\mathrm{std}(\Phi_d^{\mathrm{v}})+\varepsilon)$, ε=10^{-12}. Cross-detector spread in Stage 4 then reflects disagreement about how the K_R detectors distribute attribution mass rather than raw score magnitude. The surrogate is faithful for most detectors (median validation R^2=0.94; eight of ten exceed 0.85), with LOF and KNN the exceptions (R^2=$0.25, 0.63$). Crucially, surrogate fidelity affects *only* the attribution gate: the anchor's detection flag $f^{(a)}$ (Stage 5) is computed from the detector's *raw* score s_a, not the surrogate, and σ averages over K_R members, so a single low-fidelity surrogate cannot dominate the gate.

F. Stage 4 – The SHAP-σ Attribution-Evidence Gate

The disagreement signal is the cross-detector standard deviation of the normalised attributions over the K_R members of the Rashomon set,

$$\sigma_{ij} \;=\; \mathrm{std}_{d\in\mathcal{R}}\big(\tilde{\Phi}_{d,ij}\big). \tag{4}$$

We threshold per feature: τ_j is the empirical q-quantile of $\{\sigma_{ij}^{\mathrm{v}}\}_{i=1}^{N_{\mathrm{v}}}$ with q=0.75, and $\phi_{ij} = \mathbf{1}[\sigma_{ij}^{\mathrm{te}} > \tau_j]$. Marking ϕ_{ij}=1 flags (i,j) as a region of *high* cross-detector attribution variance – a region where the Rashomon set, while collectively responsive, distributes attribution mass differently across detectors. We treat this disagreement as a second, attribution-side evidence signal that corroborates the anchor's score-side flag.

G. Stage 5 – Anchor Flag Generation and σ-Gated Output

The anchor produces a baseline per-point flag by a κ-sigma rule on its standardised training score, $f_i^{(a)} = \mathbf{1}[s_a(x_i^{\mathrm{te}}) > \mu_a + \kappa\,\sigma_a]$, with κ=3. The canonical per-feature prediction matrix is

$$\hat{Y}_{ij} \;=\; f_i^{(a)} \wedge \phi_{ij}, \tag{5}$$

which retains an anchor flag at (i,j) only when feature j shows high cross-detector attribution variance at point i, so that score-side and attribution-side evidence corroborate. TELCO carries per-feature labels (its 12-KPI streams), so we evaluate Eq. (5) directly via macro F_1 across KPIs. Where only point-level labels are available the rule collapses naturally to $\hat{y}_i = f_i^{(a)} \wedge \phi_i^{\vee}$ with $\phi_i^{\vee} = \bigvee_j \phi_{ij}$, the OR of $\hat{Y}_{ij}$ across features. Eq. (5) is the only fusion rule reported: no auxiliary AND/OR variants and no per-series selection that would peek at the test labels. The rule is recall-monotone ($\hat{y}_i \le f_i^{(a)}$); the disjunctive collapse $\phi_i^{\vee}$ is p-sensitive (a fixed q is structurally more permissive as the feature count p grows), which is why we report the per-feature form on TELCO.

The complete procedure is summarised in Algorithm 1.

III. Experiments

A. Dataset

A multivariate telemetry dataset [1] from an operational mobile cellular network spanning January–July 2021, with p=12 KPIs sampled every five minutes and ground truth labelled by Network Operation Center experts. The official test phase contains $26{,}496$ timestamps per KPI; summed over the 12 streams there are $3{,}080$ anomaly-labelled points (≈1% per-KPI prevalence; Table III). [2] truncates the window to $25{,}143$ timestamps when comparing against C-MTAD-GAT.

B. Protocol

We split the training data into 80% training and 20% validation; detectors are fit on training only. Hyperparameters are self-tuned label-free on the training split (Sec. II-B). Quality q_d uses three random subspaces with $2{,}000$ uniform background samples per subspace. The SHAP-σ quantile is fixed at q=0.75 and the anchor κ-sigma at κ=3.

a) Evaluation metric: We report the *per-KPI macro* F_1 together with the corresponding macro precision and recall: each metric is computed independently on each of the 12 KPI binary label streams and then averaged across streams. For the deep baselines DC-VAE [1] and C-MTAD-GAT [2] we quote the per-KPI macro scores reported in their papers under this same aggregation; we do not re-train them.

IV. Results

A. Self-tuned Singleton Baselines

Table I reports the per-KPI macro F_1, precision, and recall of every self-tuned singleton detector on TELCO when its κ-sigma flag is broadcast across the 12 KPIs. Self-tuned LOF leads at F_1=0.419, narrowly above ECOD (0.401); all remaining detectors stay below 0.12. The strongest singletons LOF and ECOD differ qualitatively – LOF is distance-based

Algorithm 1 SHAP-σ with joint Rashomon-set selection

Require: $X_{\mathrm{tr}}, X_{\mathrm{v}}, X_{\mathrm{te}}$; detector pool $\mathcal{D}$ (K=10); set size K_R; strategy $\mathcal{S}\in\{$Quality, Diversity, Mixed$\}$; κ, q.
Ensure: Predicted anomaly flags $\hat{y} \in \{0,1\}^{N_{\mathrm{te}}}$.

```
Fit every d∈D on X_tr; score all splits.
Compute MV proxy A_d via Eq. (1) and quality q_d via Eq. (2).
Compute Spearman matrix ρ_ij on {s_d(X_tr)}.
Default: if (K_R, S) unspecified, set K_R=2, S=Mixed (the closed-form default; Sec. II-D).
Select R, |R|=K_R, by S (A: arg max q̄, B: arg max δ̄, C: arg max q̄δ̄; for K_R=2 C is Eq. (3)).
Anchor a ← arg max_{d∈R} q_d.
for all d ∈ R do
    Fit LightGBM g_d on (X_tr, s_d(X_tr)); compute TreeSHAP Φ_d^v, Φ_d^te; divide by std(Φ_d^v) + ε.
end for
Compute σ_ij via Eq. (4). τ_j ← q-quantile of σ_·j^v;  φ_ij ← 1[σ_ij^te > τ_j].
Compute anchor κ-sigma flag f_i^(a).
Per-feature prediction Ŷ_ij ← f_i^(a) ∧ φ_ij (Eq. (5)); for point-level evaluation return ŷ_i ← ⋁_{j=1}^p Ŷ_ij = f_i^(a) ∧ φ_i^∨.
```

TABLE I
TELCO PER-KPI MACRO METRICS OF THE 10 SELF-TUNED SINGLETONS. *Tune*: THE LABEL-FREE PARAMETER CHOICE; BOLD = BEST PER COLUMN.

Detector	F_1	P	R	Tune
LOF	**0.419**	0.460	**0.452**	k = elbow
ECOD	0.401	**0.488**	0.396	param-free
KNN	0.115	0.209	0.089	k = elbow
GMM	0.112	0.478	0.066	BIC
AutoEncoder	0.098	0.130	0.089	[64, 32], 50 ep
OCSVM	0.092	0.105	0.099	ν=0.05
IF	0.058	0.047	0.092	path-len. plateau
COPOD	0.056	0.052	0.072	param-free
PCA	0.042	0.035	0.067	95% var
HBOS	0.033	0.349	0.017	Sturges

with recall 0.452, while ECOD's per-feature CDF favours precision (0.488) – which makes them natural pair candidates downstream.

B. Joint Pair Score and Rashomon-Set Sweep

Figure 1 visualises all candidate detector pairs at K_R=2 under the closed-form joint pair score (Eq. (3)); its maximiser is LOF+ECOD, the two highest-quality singletons. The closed-form pair is a label-free heuristic: it favours quality and decorrelation, and does not in general coincide with the (label-aware) F_1-oracle pair – on TELCO the oracle pair is ECOD+HBOS – which is exactly why the pipeline adds further partners. As K_R grows under Mixed joint-subset selection, the Rashomon set absorbs additional self-tuned detectors and the pipeline F_1 climbs from 0.384 at K_R=2 into a *plateau* of 0.45–0.46 that holds for every $K_R \geq 3$ (Table II); the label-free variance-stability rule (Sec. II-D) lands in this plateau, selecting five detectors at F_1=0.460 (Table III) without consulting labels. The anchor is LOF on every K_R (the highest-quality member, q_{LOF}=1), while ECOD and HBOS enter as the first partners; this illustrates that the σ gate corroborates the anchor's score-side evidence with detectors of a different inductive bias (distance vs. empirical CDF and histogram). Adding the remaining low-quality detectors leaves

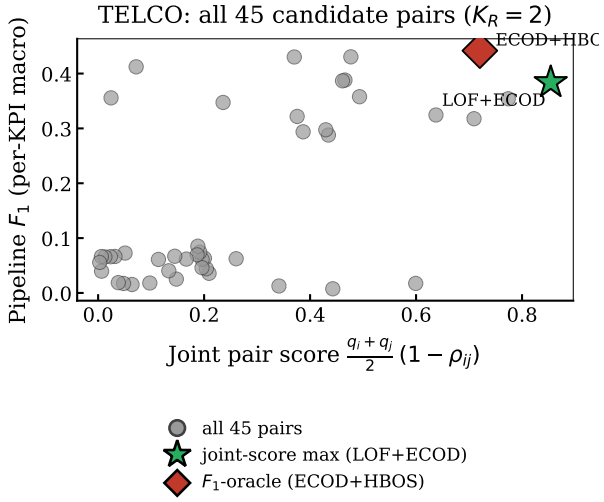


Fig. 1. Joint pair score (Eq. (3)) vs. pipeline F_1 for all K_R=2 pairs on TELCO. The label-free maximiser (green star, LOF+ECOD) differs from the F_1-oracle (ECOD+HBOS); the K_R sweep closes the gap by adding partners.

F_1 essentially flat, as they contribute only marginal cross-detector disagreement at the per-KPI scale.

C. Per-KPI Breakdown

Table III dissects the label-free quality-rule run per KPI. SHAP-σ achieves F_1>0.7 on TS3, TS5, and TS6 (well-behaved short-burst anomaly types), maintains F_1>0.55 on three further KPIs (TS4, TS7, TS8), and fails (F_1=0) only on the two near-degenerate KPIs (TS11, TS12) that contain < 25 ground-truth anomalies each over the test window. The macro precision of 0.597 implies that roughly three in five flagged points lie inside a ground-truth anomaly segment.

D. Comparison to Deep Baselines

Table IV compares SHAP-σ at its label-free operating point against the deep baselines on the per-KPI macro F_1 used by [2]. Against the values those papers report under this

TABLE II

TELCO F_1 SENSITIVITY TO THE RASHOMON-SET SIZE K_R UNDER MIXED JOINT-SUBSET SELECTION. F_1 PLATEAUS AT 0.45–0.46 FOR EVERY $K_R \geq 3$; THE LABEL-FREE QUALITY RULE (SEC. II-D) INSTEAD FIXES THE SET DIRECTLY AND LANDS IN THE SAME BAND (0.460, TABLE III). ANCHOR IS LOF THROUGHOUT; SETS NESTED. BOLD = BEST PER COLUMN.

K_R	Selected members (added on each row)	F_1	P	R
2	LOF+ECOD	0.384	0.567	0.315
3	+HBOS	0.458	0.588	0.416
4	+OCSVM	0.461	**0.600**	0.418
5	+COPOD	0.445	0.596	0.393
6	+GMM	0.461	0.594	0.430
7	+PCA	**0.462**	0.598	0.430
8	+IF	0.454	0.587	**0.431**
9	+KNN	0.453	0.586	**0.431**
10	+AE	0.452	0.584	0.430

TABLE III

PER-KPI BREAKDOWN OF THE LABEL-FREE QUALITY-RULE SET (ANCHOR LOF + ECOD, HBOS, COPOD, PCA). #TRUE/#PRED: LABELLED/FLAGGED COUNTS; TP/FP/FN: CONFUSION COUNTS.

KPI	#true	#pred	TP	FP	FN	F_1	P	R
TS1	412	162	119	43	293	0.415	0.735	0.289
TS2	501	146	112	34	389	0.346	0.767	0.224
TS3	138	162	120	42	18	**0.800**	0.741	**0.870**
TS4	179	128	95	33	84	0.619	0.742	0.531
TS5	106	125	90	35	16	0.779	0.720	0.849
TS6	107	137	88	49	19	0.721	0.642	0.822
TS7	269	197	135	62	134	0.579	0.685	0.502
TS8	267	168	132	36	135	0.607	**0.786**	0.494
TS9	564	184	115	69	449	0.307	0.625	0.204
TS10	501	162	116	46	385	0.350	0.716	0.232
TS11	21	97	0	97	21	0.000	0.000	0.000
TS12	15	55	0	55	15	0.000	0.000	0.000
macro	3,080	1,723				**0.460**	**0.597**	0.418

aggregation, SHAP-σ is 3.4× above DC-VAE (F_1=0.134 [1]) and 2.9× above C-MTAD-GAT (F_1=0.160 [2]). As these baselines are quoted rather than re-trained here, the multiplier should be read as indicative; the controlled comparison is against the best self-tuned singleton (Table I). The deep baselines also emit *global* per-timestamp alarms, so a per-KPI macro F_1 – which rewards feature-level localisation – structurally understates them; SHAP-σ's contribution is precisely to localise a global detector's alarms to the responsible KPI, which these baselines do not target.

E. Where the Gain Comes From

We attribute the improvement honestly. The bulk of it over the deep baselines comes from the label-free self-tuning and Rashomon-subset selection rather than from the σ gate alone: a single self-tuned LOF already reaches 0.419. In a controlled five-seed comparison at the operating set, the SHAP-σ gate (F_1=0.447±0.013) is on par with a naive k-of-K ensemble vote (vote≥ 2: 0.437±0.030) and does not significantly exceed it. Its advantage is therefore not raw F_1 but (i) *stability* – about half the seed variance of voting, while vote≥ 3 collapses to 0.10 – and (ii) *interpretability*: unlike a black-box vote, the gate attaches an explicit per-KPI attribution to every surviving flag, identifying *which* KPI the retained detectors agree on. The per-seed numbers are released with the code.

TABLE IV

TELCO HEADLINE: DEEP BASELINES, THE 10 SELF-TUNED SINGLETONS, AND SHAP-σ AT ITS LABEL-FREE OPERATING POINT. BOLD = BEST PER COLUMN; DEEP-BASELINE P/R ARE THE MACRO-POINTWISE VALUES FROM THE CITED PAPERS (SAME AGGREGATION).

Method	F_1	P	R
DC-VAE [1]	0.134	0.220	0.096
C-MTAD-GAT [2]	0.160	0.319	0.107
self-tuned LOF (best singleton)	0.419	0.460	**0.452**
self-tuned ECOD	0.401	0.488	0.396
self-tuned KNN	0.115	0.209	0.089
self-tuned GMM	0.112	0.478	0.066
self-tuned AutoEncoder	0.098	0.130	0.089
self-tuned OCSVM	0.092	0.105	0.099
self-tuned IF	0.058	0.047	0.092
self-tuned COPOD	0.056	0.052	0.072
self-tuned PCA	0.042	0.035	0.067
self-tuned HBOS	0.033	0.349	0.017
SHAP-σ (label-free rule)	**0.460**	**0.597**	0.418

V. CONCLUSIONS

We presented SHAP-σ, a label-free pipeline for multivariate cellular network anomaly detection that promotes the cross-detector standard deviation of SHAP attributions from a post-hoc explanation into a first-class attribution-evidence signal. The methodological contributions are: (i) label-free self-tuning of a K=10 unsupervised detector pool (including an AutoEncoder and OCSVM alongside the eight standard PyOD detectors); (ii) a closed-form Rashomon pair score $\frac{q_i+q_j}{2}(1-\rho_{ij})$ combining a label-free Mass–Volume quality proxy with Spearman score-decorrelation; and (iii) a per-KPI σ gate that filters anchor flags by the cross-detector spread of TreeSHAP attributions. On the TELCO cellular benchmark a label-free variance-stability rule selects a five-detector Rashomon set (anchor LOF), reaching per-KPI macro F_1=0.460 (precision 0.597, recall 0.418), +0.041 over the best self-tuned singleton (F_1=0.419); the result is seed-stable (0.447±0.013) and needs no label-based size selection (F_1 holds across all $K_R \geq 3$). We are explicit that this headline gain comes chiefly from the label-free self-tuning and subset selection: the σ gate is statistically on par with a naive ensemble vote, and its contribution is not raw F_1 but seed-stability and – unlike a black-box vote – explicit per-KPI attribution evidence for every flag. Against the per-KPI macro F_1 those papers report, it exceeds the deep baselines DC-VAE and C-MTAD-GAT [1], [2] by 2.9–3.4×, while remaining transparent and reproducible per-KPI.

a) Limitations and future work.: By construction the gate is recall-monotone ($\hat{Y}_{ij} \leq f_i^{(a)}$): it can only suppress anchor flags, so SHAP-σ is a *precision-first* filter, not a recall booster. We regard this as appropriate for the target setting – in operational KPI monitoring the dominant cost is alert fatigue from false positives rather than marginal recall – but it does cap recall at the anchor's, and pairing the gate with a higher-recall anchor is left to future work. The attribution step also assumes the LightGBM surrogate tracks each detector (median R^2=0.94, but only 0.25 for LOF); the gate tolerates this by

averaging over the set, yet model-native attributions would be preferable where available. The disjunctive feature-wise gate is most informative on per-feature labels (as in TELCO) and degenerates under OR-aggregation for point-level evaluation, deferred to a journal extension. We also plan to learn $(K_R, \mathcal{S})$ label-free from synthetic anomaly injection [8], [10], to compare against deep label-free baselines such as MTGFlow [33], and to extend σ-SHAP to cross-window variance for streaming deployment.

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